

George Sleen

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EDUCATION

University of British Columbia

Vancouver, BC

Bachelor's of Applied Science, Engineering Physics (Electrical Option)

2023 — 2028

- Relevant Coursework: Circuit Analysis, Signals and Systems, Motor Design, Principles of Software Construction, Instrument Design, Experimental Techniques, Electricity and Magnetism, Quantum Mechanics, Vector Calculus, Linear Algebra

WORK EXPERIENCE

Memory IO Firmware Intern

May 2026 — Present

AMD

Vancouver, BC

- Developed extremely performant firmware for AMD's HBM5 PHY
- Contributed to internal tooling for pre-silicon verification of code features in CI
- Characterized code performance and ensured that architecture targets were met

Photonics Lab Assistant

May 2026 — Present

UBC SiEPIC

Vancouver, BC

- Created the control solution for holding laser interference at quadrature in on chip MZIs
- Ported simulations from Lumerical to Python for the on chip variable optical attenuators
- Characterized a silicon photonic chip and supporting circuitry for measuring laser linewidth on chip

Electronics Design Intern

Jan 2025 — May 2025

Blue Robotics

Victoria, BC

- Designed high speed PCBs with signals up to 125 MHz
- Analyzed transmission line characteristics of custom cables and PCB traces
- Validated circuit design with SPICE tools and development PCBs
- Characterized transmission speeds and bandwidth using a Raspberry Pi and Linux

PROJECTS

Autonomous Pet Retrieval Robot (github.com/enph253-2025-team5)

May 2025

- Designed motor drivers to actuate and drive our robots (github.com/enph253-2025-team5/dual-h-bridge-pcb)
- Used FreeRTOS to handle concurrent control of actuators, sensors, and I2C buses
- Closed motor control loops with PID control, enabling centimeter level positional accuracy for our robot
- Detected specific objects using multiple I2C LiDAR modules and convolution with the expected shape

EXTRACURRICULARS

Electrical Drive Member

Sept 2025 — Present

UBC Sailbot

Vancouver, BC

- Designed an IMU for use on an autonomous Sailboat (github.com/UBCSailbot/PLRS-IMU)
- Used an Extended Kalman Filter to fuse sensor data from a MEMS IMU and dual antenna GNSS kit
- Developed simulator for HIL testing, rapid tuning iteration and functional verification of our IMU

Design Team Captain

Aug 2024 — Aug 2025

SUBC

Vancouver, BC

- Managed a team of over 60 people in the design and manufacturing of a race submarine for international competition
- Oversaw a budget of \$40,000

SKILLS

Software/Firmware: Git, C/C++, FreeRTOS, Agentic AI, Python, Bash, GDB, Linux, Controls, Rust, Docker, OOP, RTOS, Scripting, CI/CD, ROS, Networking, Computer Vision, Machine Learning, Nix, Simulation, Profiling Tools, HIL

Electrical: Circuit Analysis, SPICE, KiCad, Soldering, Transmission Lines, Oscilloscopes, SystemVerilog, Digital Logic, Computer Architecture, Altium, SWD, JTAG, Controls, STM32, Registers, DMA, Motors, HIL Testing, State Estimation